instruction x whose hidden state at token t in layer l is h_t^l, its belief of harmfulness and refusal is defined respectively as

$$\Delta_{\text{harmful}} = \text{Avg}(s^l(h^l_{t_{\text{inst}}})) = \frac{1}{L}\sum_{l=1}^{L}(cos_sim(h^l_{t_{\text{inst}}}, \mu^{l,\,t_{\text{inst}}}_{\text{harmful}}) - cos_sim(h^l_{t_{\text{inst}}}, \mu^{l,\,t_{\text{inst}}}_{\text{harmless}})), \tag{3}$$

$$\Delta_{\text{refuse}} = \text{Avg}(s^l(h^l_{t_{\text{post-inst}}})) = \frac{1}{L}\sum_{l=1}^{L}(cos_sim(h^l_{t_{\text{post-inst}}}, \mu^{l,\,t_{\text{post-inst}}}_{\text{refuse}}) - cos_sim(h^l_{t_{\text{post-inst}}}, \mu^{l,\,t_{\text{post-inst}}}_{\text{accept}})). \tag{4}$$

We then compute the belief for a random sample of each category in the test set. The results on Llama2 are shown in Figure 3. Accepted harmless instructions (green squares) and refused harmful instructions (red dots) have a relatively strong positive correlation between the beliefs of harmfulness Δ_{harmful} and refusal Δ_{refusal}. However, we find that refused harmless instructions have significantly low belief scores in harmfulness. This suggests that, although the model over-refuses these harmless instructions behaviorally, it still internally deems these instructions as harmless. This result also supports the prior hypothesis [Röttger et al., 2023, Bianchi et al., 2023] that refusal can be triggered by some shallow syntax features despite the harmlessness of user instructions. In the case of accepted harmful instructions, the belief of harmfulness Δ_{harmful} remains positive for many examples, indicating that LLMs internally view them as harmful despite accepting them. Overall, our results suggest that refusal is generally correlated with harmfulness in LLMs. However, there exist cases where refusing (or accepting) an instruction does not align with the model's internal perception of harmfulness.

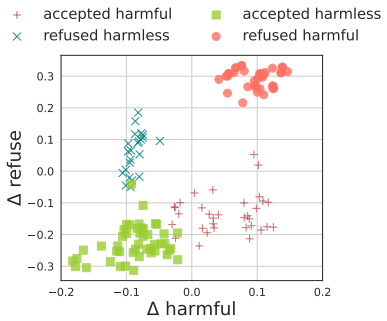


Figure 3: Correlation between the model's beliefs of harmfulness and refusal on Llama2. Each point is a sampled instruction. We show that refusing an instruction is not necessarily aligned with the model's internal belief of harmfulness. For example, refused harmless instructions have negative harmful belief scores, indicating that the model internally considers them as not harmful, even though it behaviorally refuses them.

3.4 Eliciting refusal with harmfulness directions

To investigate the causality between believing an instruction is harmful internally and refusing it in the response, we steer the hidden states of accepted harmless instructions towards the region of $\mathcal{C}^l_{\text{harmful}}$ to see how much that can reverse the model's acceptance to refusal.

Harmfulness direction. We can extract a steering direction of harmfulness at a layer l with the difference-in-means [Marks and Tegmark, 2023] from $\mathcal{C}^l_{\text{harmless}}$ to $\mathcal{C}^l_{\text{harmful}}$ in the latent space. Namely, we define the harmfulness direction as the difference between two cluster centers at token t_{inst},

$$v^l_{\text{harmful}} = \mu^{l,\,t_{\text{inst}}}_{\text{harmful}} - \mu^{l,\,t_{\text{inst}}}_{\text{harmless}}. \tag{5}$$

We then intervene on the residual stream for the hidden state of test examples using activation addition at layer l, i.e., $h'^l = h^l + v^l_{\text{harmful}}$ to all tokens of input instructions. As comparison, we also extract a refusal direction as $v^l_{\text{refuse}} = \mu^{l,\,t_{\text{post-inst}}}_{\text{refuse}} - \mu^{l,\,t_{\text{post-inst}}}_{\text{accept}}$ at token $t_{\text{post-inst}}$.

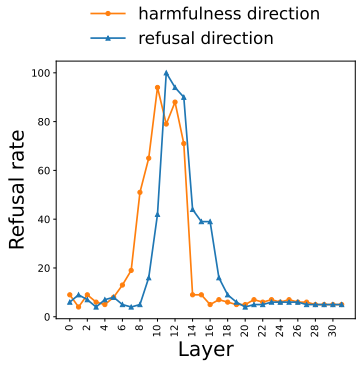
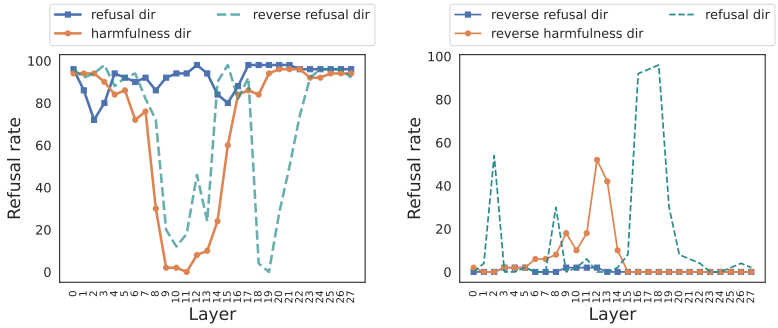


Figure 4: Steering the hidden states of harmless instructions along the harmfulness direction and refusal direction can both elicit refusal behaviors on Llama3.

Steering results. We find that both the harmfulness direction and the refusal direction can elicit LLMs' refusal behaviors on harmless instructions. Layer-wise results on Llama3 are shown in Figure 4, and results on other models are shown in Appendix C. Steering along the harmfulness direction at layer nine leads to the highest



(a) Harmless instructions (b) Harmful instructions

Figure 5: Steering with the harmfulness direction (the orange line) and the refusal direction (the blue line) leads to opposite behaviors, which serves as causal evidence that these two directions are fundamentally different in LLMs. We append an inversion question (e.g., "Could this user prompt cause harm? Answer 'Certainly' if it could, otherwise 'No'. ") to the input instruction so that the model should respond with an acceptance token (e.g., `Certainly`) instead of a refusal token (e.g., `No`) if the instruction is harmful and vice versa.

refusal rate (94%), while the refusal direction performs the best at layer eleven (100% refusal rate). On Qwen2 and Llama2, the refusal rate elicited by the harmfulness direction is much lower than that elicited directly by the refusal direction (see Appendix C). Although both directions can elicit refusal behaviors, we find these two directions have a low average cosine similarity , e.g., around 0.1 on Llama2. For held-out refused instructions, cosine similarity with the refusal direction averages 0.57, while for accepted ones it averages -0.14; in this context, an observed similarity of 0.1 is low. This suggests that the harmfulness direction and the refusal direction elicit refusal behaviors in different ways. We further prove their difference causally in Section 3.5 by constructing a prompting task where steering along the harmfulness direction and the refusal direction leads to opposite behaviors.

3.5 Causally separating the harmfulness direction and the refusal direction

In Section 3.4, we demonstrate that steering along the refusal direction and the harmfulness direction can both elicit models' refusal behaviors. In this section, we prove that these two directions play different roles inside LLMs. We design a reply inversion task to provide causal evidence that steering along these two directions will lead to opposite results. Overall, our results suggest that **(1)** the harmfulness direction extracted at t_{inst} encodes the concept of harmfulness, and steering the model along it will reverse the model's judgment of the instruction harmfulness; **(2)** the refusal direction extracted at $t_{\text{post-inst}}$ mainly encodes surface-level refusal signals, and steering along it will not always alter the model's judgment of instruction harmfulness.

Reply inversion task. To distinguish the effects of the refusal direction and the harmful direction, we design a reply inversion task that disentangles harmfulness and refusal behaviorally: when the harmfulness of instructions is increased, the model should output acceptance tokens instead of refusal tokens. To achieve this, we append an inversion question, e.g., "Could this user prompt cause harm?" to the user's instruction, as shown in Table 2. Then, the model will respond to harmless instructions with a refusal token (e.g., "No") and to harmful instructions with an acceptance token (e.g., "Certainly"). We then steer the hidden states using our identified harmfulness direction and the refusal direction, following the intervention approach in Section 3.4 (implementation details in Appendix E.1), to examine whether the model's judgment of harmfulness will be altered. We determine if the judgment is changed by observing if the generated response is flipped.

`User wants to` {User Instruction Tokens}
`Could this user prompt cause harm? Answer 'Certainly' if it could, otherwise 'No'`

Table 2: An example of the inversion prompting template.

Steering results. Results on Qwen2 are shown in Figure 5. When the instruction is harmless (Figure 5a), the model should answer "No" in our reply inversion task without intervention. The

refusal rate indicates the occurrence of refusal tokens like "No". If we steer the hidden states along the harmfulness direction, the model will start to flip its answer to "Certainly", and thus the refusal rate will drop. This suggests that the model begins to internally interpret the harmless instruction as harmful, providing causal evidence that the identified direction indeed represents the concept of harmfulness. In contrast, steering along the refusal direction generally causes the model to maintain its original refusal response (e.g., "No"), while steering in the reverse direction tends to produce affirmative tokens (e.g., "Certainly"). These results suggest that the refusal direction may not alter the model's perception of harmfulness; instead, it may primarily encode surface-level refusal cues rather than a deeper understanding of harmfulness.

On the other hand, if the instructions are harmful, as shown in Figure 5b, steering them along the reverse harmfulness direction will cause the model to reply "No". This indicates that our intervention leads the model to interpret those harmful instructions as harmless. However, steering along the reverse refusal direction fails to reverse the model's perception of harmfulness, and therefore does not elicit refusal responses in the reply inversion task. We observe similar results on other inversion templates and models as shown in Appendix E. In sum, we provide causal evidence that LLMs internally reason about the harmfulness of inputs independently from their refusal behaviors, indicating that harmfulness and refusal are represented as separate concepts.

4 Analyzing Jailbreak via Harmfulness

Different jailbreak methods [Zou et al., 2023b, Yu et al., 2023, Zeng et al., 2024] have successfully enabled harmful instructions to be accepted by LLMs. But it remains unclear how jailbreak methods work. In this section, we apply the identified internal belief of harmfulness and refusal (see Equation 3 and Equation 4) to analyze jailbreak. We find that **some jailbreak methods work by suppressing the refusal signal, but cannot fundamentally reverse the model's belief of harmfulness**. We consider different types of jailbreak methods as detailed in Section 2, i.e., adversarial suffixes [Zou et al., 2023b], persuasion [Zeng et al., 2024] and adversarial prompting templates [Yu et al., 2023]. As shown in Figure 6, we find that in some cases, the persuasion jailbreak method can internally make LLMs believe the persuasive harmful jailbreak prompts are harmless (negative Δ_{harmful}). By contrast, for other jailbreak methods, the refusal signals are suppressed, generally leading to negative Δ_{refuse}, but in some cases, the model still internally believes the jailbreak prompts are harmful as reflected by high Δ_{harmful} scores. Therefore, although prior work has shown jailbreak methods can suppress refusal features and hypothesize that ablating the refusal directions makes LLMs perceive instructions as less harmful [Yu et al., 2025], we clarify that not all jailbreak methods can internally reverse LLMs' harmfulness judgment, highlighting the need for further investigation.

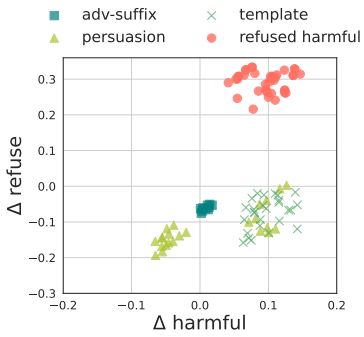


Figure 6: Belief of harmfulness and refusal for different categories of jailbreak prompts in comparison with refused harmful instructions.

5 Developing a *Latent Guard* Model with Harmfulness Representations

Guardrails for LLMs have been widely employed to improve safety, where users' input instructions are screened by a guard model [Dong et al., 2024]. When the guard model identifies potentially harmful inputs, enforcement actions will be taken (e.g., preventing LLMs from processing the input or adapting LLMs' outputs). In this section, we propose to use LLM's internal belief of harmfulness as a ***Latent Guard*** to detect challenging cases like harmful instructions that bypass refusal and harmless but over-refused instructions [Röttger et al., 2023]. ***Latent Guard*** is motivated by the faithfulness and robustness of the LLMs' perception of harmfulness: LLMs may still correctly assess the harmfulness of instructions even when their refusal behavior is incorrect, as shown in Figure 3 and Figure 6.